

Important Questions

What is Computer, why it is important?

What is Internet, What is www?

Difference between ram and rom?

Difference between http and https?

What is CPU?

What is Octa-core and quad-core?

What is multitasking and multiuser and multithreading multiprocessing?

Introduction To Computer

A **computer** is an **electronic device** that is capable of **receiving data (input)**, **processing** it as per given instructions (program), and then **producing output**. It can store data and perform various tasks very quickly and accurately.

Computer word is derived from the Latin word "**computare**", which means "**to calculate**".

Characteristics of a Computer:

- ▶ **Speed** - Computers can perform millions of instructions per second.
- ▶ **Accuracy** - High accuracy with minimum errors.
- ▶ **Automation** - Can work automatically once given instructions.
- ▶ **Storage** - Can store large amounts of data permanently.
- ▶ **Versatility** - Can perform different types of tasks.
- ▶ **Diligence** - Unlike humans, computers do not get tired or bored.

Types Of Computer

a) Analog Computer:

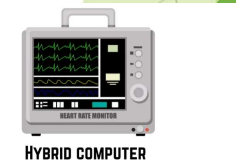
- ▶ Works on continuous data (not digital).
- ▶ Used in scientific and engineering applications.
- ▶ Example: Speedometer, thermometer.

b) Digital Computer:

- ▶ Works on discrete data (0 and 1 - binary).
- ▶ Most common type of computer used today.
- ▶ Example: Laptop, mobile phone.

c) Hybrid Computer:

- ▶ Combination of analog and digital computer.
- ▶ Used in hospitals (e.g., ECG machines), scientific applications



Types Of Digital Computer

a) Microcomputer (Personal Computer - PC):

- ▶ Smallest and most commonly used.
- ▶ Designed for individual use.
- ▶ Example: Desktop, Laptop.

b) Mini Computer:

- ▶ Mid-sized computer.
- ▶ Supports multiple users at the same time.
- ▶ Used in small and medium organizations.

c) Mainframe Computer:

- ▶ Large and powerful systems.
- ▶ Can support hundreds or thousands of users simultaneously.
- ▶ Used in banks, railways, government departments.

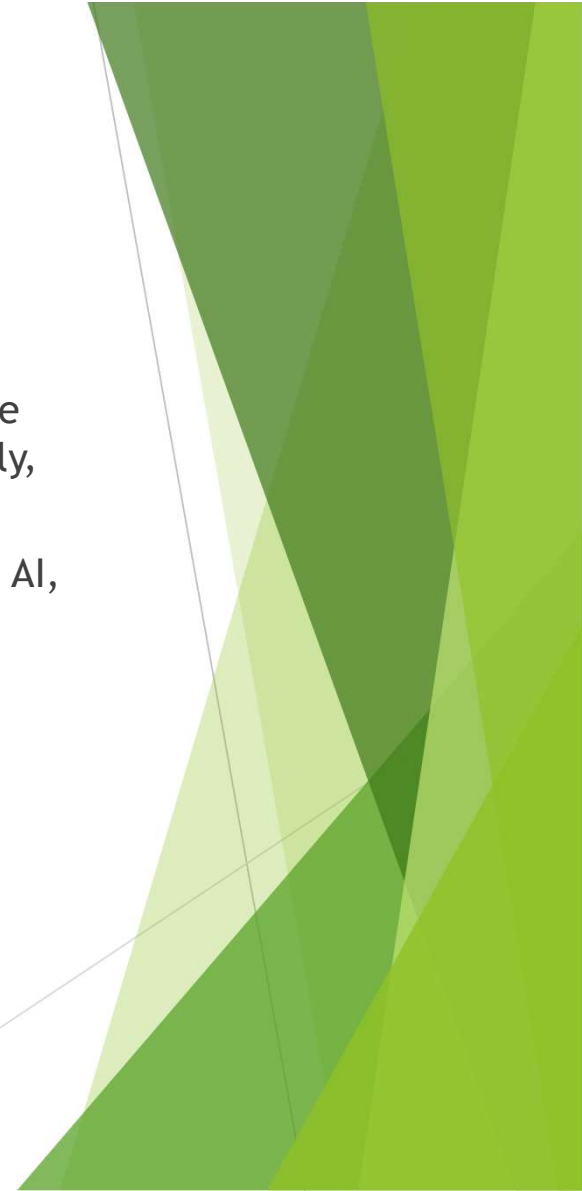
d) Supercomputer:

- ▶ Most powerful and fastest computer.
- ▶ Used in scientific research, weather forecasting, space research.
- ▶ Example: PARAM, Fugaku.



What are IT Gadgets?

- ▶ **IT (Information Technology) gadgets** are **smart electronic devices** that use computing technology to help users perform various digital tasks more easily, efficiently, and in a mobile or user-friendly way.
- ▶ These gadgets use internet, wireless technology, sensors, cloud computing, AI, etc., to enhance usability and features.



List of Latest IT Gadgets and Their Applications:

Smartphones:

- ▶ Mini computers with calling, texting, browsing, gaming, GPS, camera, etc.
- ▶ Applications:
- ▶ Social media, online banking, learning apps, video calls, mobile payments (UPI, Paytm), etc.

Tablets and iPads:

- ▶ Portable touchscreen devices.
- ▶ Applications:
- ▶ Reading, attending online classes, entertainment, sketching, note-making, browsing.

Smartwatches:

- ▶ Wearable device that connects with smartphone.
- ▶ Applications:
- ▶ Track health (heart rate, steps), receive calls/messages, GPS tracking, fitness monitoring.

Laptops and Notebooks:

- ▶ Portable version of desktop computers.
- ▶ Applications:
- ▶ Office work, online learning, programming, browsing, designing, business tasks.

Voice Assistants (e.g., Alexa, Google Assistant, Siri):

- ▶ Use AI to understand voice commands and perform tasks.
- ▶ Applications:
- ▶ Home automation, asking questions, setting reminders, playing music, controlling smart devices.

Smart TVs:

- ▶ Television with internet connectivity and apps.
- ▶ Applications:
- ▶ Watch YouTube, Netflix, online content, connect phones, use voice control.



Drones:

- ▶ Remotely operated flying machines with cameras.
- ▶ Applications:
- ▶ Aerial photography, surveillance, agriculture, military, delivery services.

VR (Virtual Reality) Headsets:

- ▶ Devices that give immersive 3D experience.
- ▶ Applications:
- ▶ Gaming, training simulations, education, architecture, entertainment.

Smart Speakers:

- ▶ Wireless speaker with virtual assistant.
- ▶ Applications:
- ▶ Voice-controlled music, news updates, controlling smart home devices.



Smart Home Devices (IoT Devices):

- ▶ Internet-connected devices used at home.
- ▶ Examples: Smart bulbs, smart locks, smart ACs, smart refrigerators.
- ▶ Applications:
- ▶ Home automation, remote control via mobile apps.

Applications of Computers and IT Gadgets in Daily Life:

Area

Education

Healthcare

Banking

Business

Entertainment

Travel and Transport

Government Services

Applications

Online classes, e-books, virtual labs

Patient monitoring, telemedicine, digital reports

Online banking, ATMs, digital payments

Data processing, emails, video meetings

Music, movies, online games, social media

Online bookings, GPS navigation, smart cards

Online forms, public portals, Aadhaar verification

Evolution of Computers (History)

- ▶ The **evolution of computers** refers to the development of computing machines from simple calculating tools to the highly intelligent systems we use today.
- ▶ The journey of computers began with basic mechanical devices and moved forward through various stages of technological advancement.



Early Calculating Devices

Device	Inventor	Year	Description
Abacus	Chinese Civilization	~3000 BC	First manual calculator using beads.
Napier's Bones	John Napier	1617	Used for multiplication and division.
Pascaline	Blaise Pascal	1642	First mechanical calculator.
Analytical Engine	Charles Babbage	1837	Considered the first concept of a modern computer.

Generations of Computers

- ▶ The development of computer technology is divided into **five generations**, each marked by significant advancements in hardware and software.
- ▶ **1st Generation Computers (1940-1956)**
- ▶ **Technology Used:** Vacuum Tubes
- ▶ **Programming Language:** Machine language (binary code - 0s and 1s)
- ▶ **Size:** Very large (occupied entire rooms)
- ▶ **Speed:** Very slow
- ▶ **Examples:** ENIAC, EDVAC, UNIVAC
- ▶ **Drawbacks:**
 - ▶ Produced a lot of heat
 - ▶ Required air conditioning
 - ▶ Frequently failed

- ▶ **2nd Generation Computers (1956-1963)**
- ▶ **Technology Used:** Transistors (replaced vacuum tubes)
- ▶ **Programming Language:** Assembly language
- ▶ **Size:** Smaller and more reliable
- ▶ **Speed:** Faster than 1st generation
- ▶ **Examples:** IBM 1401, IBM 7094
- ▶ **Advantages:**
 - Less heat
 - Less power consumption
 - Better performance



- ▶ **3rd Generation Computers (1964-1971)**
- ▶ **Technology Used:** Integrated Circuits (IC)
- ▶ **Programming Language:** High-level languages (like FORTRAN, COBOL)
- ▶ **Size:** More compact
- ▶ **Speed:** Much faster
- ▶ **Examples:** IBM System/360
- ▶ **Advantages:**
 - ▶ Lower cost
 - ▶ Better storage and processing
 - ▶ Supported multitasking

- ▶ **4th Generation Computers (1971-Present)**
- ▶ **Technology Used:** Microprocessors (first was Intel 4004)
- ▶ **Programming Language:** All modern programming languages
- ▶ **Size:** Very small - personal computers
- ▶ **Speed:** Very high processing speed
- ▶ **Examples:** Desktop computers, laptops, tablets
- ▶ **Advantages:**
 - ▶ Affordable for personal use
 - ▶ Highly reliable
 - ▶ Use of GUI (Graphical User Interface), mouse, keyboard

- ▶ **5th Generation Computers (Present & Beyond)**
- ▶ **Technology Used:** Artificial Intelligence (AI), robotics, cloud computing, quantum computing
- ▶ **Programming Concepts:** Machine learning, Natural Language Processing (NLP)
- ▶ **Features:**
 - ▶ Self-learning systems
 - ▶ Voice and image recognition
 - ▶ Real-time processing
- ▶ **Examples:** AI assistants (Alexa, Siri), smart robots, quantum computers
- ▶ **Advantages:**
 - ▶ Intelligent decision making
 - ▶ Automation of complex tasks
 - ▶ IoT (Internet of Things) integration

What is CPU?

- ▶ The **CPU (Central Processing Unit)** is known as the “**Brain of the Computer**”. It is responsible for performing all calculations, processing data, and controlling all the operations of the computer system.
- ▶ It processes the instructions given by the user and executes them.
- ▶ **Major Functions of CPU:**
- ▶ **Processing Data:**
It processes the data according to the given instructions.
- ▶ **Controlling Operations:**
It controls the working of all parts of the computer.
- ▶ **Performing Calculations:**
It performs arithmetic and logical operations.
- ▶ **Fetching & Executing Instructions:**
It fetches instructions from memory and executes them step-by-step.

Main Components of CPU

The CPU is mainly divided into **three parts**:

ALU (Arithmetic and Logic Unit)

- ▶ Performs all **arithmetic operations** like addition, subtraction, multiplication, division.
- ▶ Performs **logical operations** like comparison (greater than, less than, equal to).
- ▶ It is the **calculating unit** of the CPU.

CU (Control Unit)

- ▶ Manages and controls all activities inside the computer.
- ▶ It directs the flow of data between memory, ALU, and input/output devices.
- ▶ Does **not process** data itself but **controls** how other parts work.

Registers

- ▶ Small high-speed memory inside the CPU.
- ▶ Temporarily stores data, instructions, or results during processing.
- ▶ Faster than RAM and helps in **quick access** to frequently used data.

How CPU Works (In Simple Steps):

- ▶ **Fetch** - The control unit fetches instructions from memory.
- ▶ **Decode** - The instruction is decoded to understand the operation.
- ▶ **Execute** - ALU executes the instruction.
- ▶ **Store** - Result is stored in memory or sent to output.

Important Points:

- ▶ CPU is installed on the **motherboard**.
- ▶ The performance of a computer depends largely on the **speed and power of the CPU**.
- ▶ CPU speed is measured in **GHz (Gigahertz)**.
- ▶ Modern CPUs are **multi-core**, meaning multiple processing units in one chip (like Dual-core, Quad-core).

What are Input Devices?

- ▶ **Input Devices** are hardware components that are used to **enter data and instructions** into the computer system for processing.
- ▶ These devices allow the user to interact with the computer by providing raw data, which is later processed by the CPU.

Functions of Input Devices:

- ▶ Accept data from the user.
- ▶ Convert human-readable data into machine-readable form (binary).
- ▶ Send data and instructions to the CPU for processing.

Commonly Used Input Devices:

Keyboard

- ▶ Most commonly used input device.
- ▶ Used to type letters, numbers, and symbols.
- ▶ Contains keys like alphabets (A-Z), numbers (0-9), function keys (F1-F12), special keys (Enter, Shift, Ctrl).

Mouse

- ▶ A pointing device used to control the pointer on the screen.
- ▶ Used for clicking, selecting, dragging, and scrolling.
- ▶ Contains left button, right button, and scroll wheel.

Scanner

- ▶ Used to convert physical documents or images into digital form.
- ▶ Scanned files can be edited, saved, or shared easily.
- ▶ Types: Flatbed Scanner, Hand-held Scanner.

Microphone

- ▶ Used to input **audio or voice** into the computer.
- ▶ Used in voice recording, video calls, speech recognition.

Webcam

- ▶ Captures live **video or images**.
- ▶ Commonly used for video conferencing, online classes, streaming.



Joystick

- ▶ Used to play video games or control simulation software.
- ▶ Works by tilting in different directions.

Touchscreen

- ▶ A display screen that also acts as an input device.
- ▶ Users can give input by touching icons or text directly on the screen.
- ▶ Commonly used in smartphones, tablets, ATMs.

Trackball

- ▶ Similar to a mouse, but with a rotating ball on the top.
- ▶ The ball is rolled using fingers to move the cursor.
- ▶ Often used in laptops or special control systems.

Light Pen

- ▶ A pen-like device used to draw directly on a screen or select objects.
- ▶ Mostly used in design, drawing, or CAD applications.



Barcode Reader

- ▶ Used to scan barcodes on products.
- ▶ Commonly used in shopping malls, warehouses, and libraries.

Biometric Devices

- ▶ Used to input personal identification data like fingerprints, iris scan, or facial recognition.
- ▶ Used for security and authentication purposes (e.g., Aadhaar authentication).

Output Devices

- ▶ **Output Devices** are hardware components used to **display, print, or present the results** of data processing performed by the computer.
- ▶ They convert **machine-readable data** (binary) into **human-readable form** such as text, images, sound, or printed output.

Functions of Output Devices:

- ▶ Receive processed data from the CPU.
- ▶ Convert it into meaningful output.
- ▶ Present the result to the user in different forms (visual, audio, printed).
- ▶ Types of Output Devices:

Monitor (Visual Display Unit - VDU)

- ▶ Displays **visual output** from the computer (text, graphics, videos).
- ▶ Comes in different types:
 - ▶ **CRT (Cathode Ray Tube)** - Old, bulky.
 - ▶ **LCD (Liquid Crystal Display)** - Slim, energy-efficient.
 - ▶ **LED (Light Emitting Diode)** - Brighter, modern displays.
- ▶ Acts as a primary output device.

Printer

- ▶ Used to get **hard copy** (printed form) of documents or images.
- ▶ Converts digital data into printed text or graphics on paper.

Type	Description
Inkjet Printer	Sprays ink, good for home use.
Laser Printer	Uses laser technology, fast and clear.
Dot Matrix Printer	Uses pins and ink ribbon, used in bill printing.

Speakers

- ▶ Output device for **sound/audio**.
- ▶ Plays music, system sounds, videos, etc.
- ▶ Often used in multimedia systems, games, and video calls.

Headphones/Earphones

- ▶ Personal audio output devices.
- ▶ Allow only the user to hear sound privately.
- ▶ Useful for online classes, meetings, and listening to music

Projector

- ▶ Projects the output (from a monitor or computer) onto a **large screen or wall**.
- ▶ Used in classrooms, seminars, conferences.
- ▶ Helps in group presentations and public viewing.



Plotter

- ▶ Special type of printer used to print **large-sized drawings or graphics** like maps, engineering designs, architectural plans.
- ▶ Used by engineers, architects, and designers.

Braille Display (for Visually Impaired)

- ▶ Converts digital text into **Braille script** using raised dots.
- ▶ Helps visually impaired users to read digital content.



What is Memory?

Computer memory is a component that stores **data, instructions, and information** either temporarily or permanently. It plays a vital role in the overall performance of a computer.

Types of Memory

1. Primary Memory (Main Memory)

This is the main working memory of the computer. It is **directly accessible by the CPU**, faster but limited in capacity.

Types of Primary Memory:

a) RAM (Random Access Memory)

- ▶ **Volatile** memory (data is lost when power is off)
- ▶ Temporarily holds data and programs currently in use
- ▶ Allows both read and write operations
- ▶ More RAM = better multitasking
- ▶ Example: When you open an app or a browser tab, it runs in RAM.

b) ROM (Read-Only Memory)

- ▶ **Non-volatile** memory (data remains even after shutdown)
- ▶ Stores **firmware** or system instructions like the BIOS
- ▶ Data is mostly written during manufacturing
- ▶ Cannot be modified easily
- ▶ It helps in booting up the system.

Secondary Memory (Storage Devices)

- ▶ This memory is used to **store data permanently**. It is not directly accessed by the CPU – data is first transferred to RAM.
- ▶ Examples of Secondary Memory

Storage Device

HDD (Hard Disk)

SSD

CD/DVD

USB Drive

Memory Card

External HDD

Description

Magnetic disk, large capacity

Faster and more reliable than HDD

Optical media, used for data and videos

Portable and commonly used

Used in smartphones, cameras, etc.

Used for backups and large file storage

Difference Between Primary and Secondary Memory

Feature

Speed

Volatility

Storage Capacity

Cost

Access

Primary Memory (RAM/ROM)

Very fast

RAM is volatile

Limited

More expensive per GB

Direct by CPU

Secondary Memory (HDD/SSD)

Slower

Non-volatile

High

Cheaper per GB

Not direct, accessed via software

Cache Memory

- ▶ A **high-speed** memory located between CPU and RAM
- ▶ Stores frequently accessed data and instructions
- ▶ Much faster than RAM but very small in size
- ▶ Speeds up processing time

Registers

- ▶ **Smallest and fastest memory** inside the CPU
- ▶ Holds data/instructions that the CPU is currently processing
- ▶ Very limited in size

Memory Units (from smallest to largest)

Unit	Equivalent
1 Bit	Smallest unit (0 or 1)
1 Byte	8 Bits
1 KB	1024 Bytes
1 MB	1024 KB
1 GB	1024 MB
1 TB	1024 GB
1 PB	1024 TB

Volatile vs Non-Volatile Memory

Type	Description	Examples
Volatile Memory	Loses data when power is turned off	RAM, Cache
Non-Volatile	Keeps data without power	ROM, HDD, SSD

Types of RAM (Random Access Memory)

- ▶ There are mainly **two types** of RAM:

1.1 DRAM (Dynamic RAM)

- ▶ Stores each bit of data in a **capacitor**.
- ▶ Needs to be **refreshed continuously** to retain data.
- ▶ **Cheaper and slower** than SRAM.
- ▶ Used in most personal computers.

1.2 SRAM (Static RAM)

- ▶ Stores data using **flip-flops (transistors)**.
- ▶ **Does not need refreshing**, retains data as long as power is on.
- ▶ **Faster and more expensive** than DRAM.
- ▶ Used in **cache memory** (inside CPU).

Optional: Other Variants

- ▶ **SDRAM (Synchronous DRAM)** - synchronized with CPU clock.
- ▶ **DDR RAM (Double Data Rate)** - transmits data twice per clock cycle.
- ▶ DDR2, DDR3, DDR4, DDR5 are faster versions.

What is ROM?

- ▶ ROM (Read Only Memory) is a **non-volatile memory** used to store **firmware** or **system boot instructions**, i.e., data that **does not change** and is needed to start the computer.
- ▶ **Internal Function of ROM:**
- ▶ Stores **permanent data** that is written at manufacturing time.
- ▶ Helps computer perform **POST (Power-On Self-Test)** during startup.
- ▶ Loads **BIOS** which initializes hardware before handing control to OS.

Main Components of ROM:

Component

Decoder

Memory Cell Array

Address Lines

Data Lines

Control Lines

Function

Selects memory cell to be read

Holds data in 0s and 1s

Carry location information

Carry stored data to output

Control reading operations



Working of ROM:

- ▶ CPU sends **address** of the required instruction.
- ▶ ROM **decoder** activates that memory location.
- ▶ Stored **data is read** and sent to CPU via data lines.
- ▶ CPU uses that data for system startup (booting).

What is a Hard Disk?

A Hard Disk Drive (HDD) is a **non-volatile secondary storage device** that stores OS, files, applications, videos, etc., permanently.

Parts of a Hard Disk:

Part

Description

Platter

Round magnetic disks where data is stored

Spindle

Holds and spins the platters at high speed

Read/Write Head

Reads or writes data on the platters

Actuator Arm

Moves the read/write head across platters

Actuator

Controls the movement of actuator arm

Disk Controller

Interface between the hard disk and computer

Cache (Buffer)

Temporary memory inside HDD to speed up access



Working of a Hard Disk:

- ▶ When you open a file, the **CPU sends a request** to the hard disk.
- ▶ The **disk controller** receives the request and sends it to the actuator.
- ▶ The **actuator arm moves the read/write head** to the correct location on the spinning platter.
- ▶ The **read head reads** the magnetic signals and sends data back to CPU via controller.
- ▶ While saving files, **write head** converts data into magnetic signals and stores it on the platter.
- ▶ The platters spin at high speed (e.g., 5400 or 7200 RPM), allowing fast access to data.

